

13.1 Universal Gravitation

Question Paper

Course	CIEA Level Physics
Section	13. Gravitational Fields
Topic	13.1 Universal Gravitation
Difficulty	Easy

Time allowed: 40

Score: /28

Percentage: /100

Question 1a

Planets create large gravitational fields.

State what is meant by a *gravitational field*.

[2 marks]

Question 1b

The Earth has a larger gravitational field strength than the moon.

State what is meant by *gravitational field strength*.

[1 mark]

Question 1c

Saturn has a gravitational field strength of 10.5 N kg^{-1} on its surface and a radius of 58 000 km.

(i)

State the acceleration due to gravity of an object on Saturn.

[1]

(ii)

Calculate the mass of Saturn.

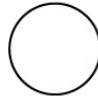
[3]

[4 marks]

Question 1d

Fig 1.1 represents Saturn as an isolated uniform sphere.

Sketch the lines to represent the gravitational field outside the sphere.



[2 marks]

Question 2a

State Newton's law of gravitation.

[2 marks]

Question 2b

Newton's law of gravitation assumes the masses are point masses.

The planets in our Solar System and the Sun are not point masses.

Explain why the law still applies to the planets in our Solar System.

[2 marks]

Question 2c

Explain why g is approximately constant for small changes in height near the Earth's surface.

[1 mark]

Question 2d

The gravitational force between the Earth and the Sun is 3.52×10^{22} N.

Calculate the radius of the Earth's orbit around the Sun.

Mass of Earth = 6.0×10^{24} kg

Mass of Sun = 2.0×10^{30} kg

[3 marks]

Question 3a

Explain what is meant by a *geostationary orbit*.

[3 marks]

Question 3b

A satellite of mass m is in a circular orbit around a planet of mass M with linear speed v .

Show that M is given by the expression:

$$M = \frac{v^2 r}{G}$$

[2 marks]

Question 3c

Show that the mass M can also be written in terms of the time period of the satellite T as

$$M = \frac{4\pi^2 r^3}{T^2 G}$$

[2 marks]**Question 3d**

Calculate the radius of a geostationary orbit around Earth.

Mass of Earth = 6.0×10^{24} kg

[4 marks]



Model Answers: Easy

1a

(a) A gravitational field is:

- A region / area / volume of space; [1 mark]
- where a mass experiences a force; [1 mark]

[Total: 2 marks]

1b

(b) Gravitational field strength is:

- The force per unit mass (on an object); [1 mark]

[Total: 1 mark]

1c

(c)

(i) The acceleration due to gravity on Saturn is:

- 10.5 m s^{-2} [1 mark]

(ii) The mass of Saturn is:

List the known quantities:

- Gravitational field strength, $g = 10.5 \text{ N kg}^{-1}$
- Radius of Saturn, $r = 58\,000 \text{ km} = 58\,000 \times 10^3 \text{ m}$
- gravitational constant, $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

Determine the equation for the mass of Saturn, M :

- $g = \frac{GM}{r^2} \Rightarrow M = \frac{gr^2}{G}$ [1 mark]

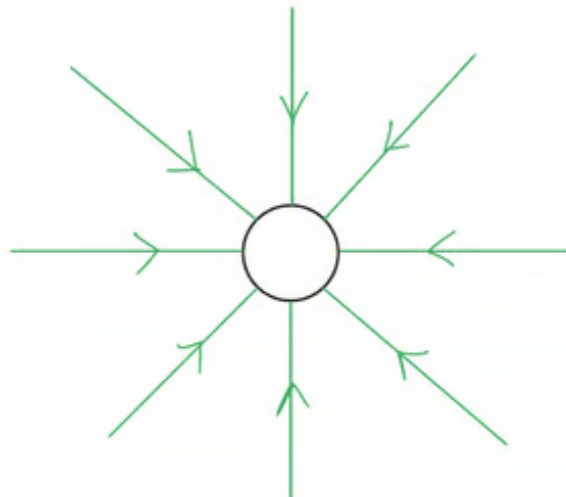
Substitute in the values:

- $M = \frac{10.5 \times (58\,000 \times 10^3)^2}{6.67 \times 10^{-11}}$ [1 mark]
- $M = 5.296 \times 10^{26} = 5.3 \times 10^{26} \text{ kg}$ [1 mark]

[Total: 4 marks]

1d

(d) The gravitational field lines around Saturn should look like:



- Lines radially outwards from the centre of the circle; [1 mark]
- Arrows pointing towards the centre of the circle; [1 mark]

[Total: 2 marks]

2a

(a) Newton's law of gravitation states that:

- The (gravitational) force is proportional to the product of the masses; [1 mark]
- And inversely proportional to the square of their separation; [1 mark]

[Total: 2 marks]

2b

(b) The law still applies to planet in our Solar System because:

- All the planets orbit the Sun; [1 mark]
- The separation of the planets (from the Sun) is much greater than the radius of the Sun / planet; [1 mark]

[Total: 2 marks]

2c

(c) g is approximately constant for small changes in height near the Earth's surface because:

- Changes in height on the Earth's surface as much smaller than the radius of the Earth; [1 mark]

[Total: 1 mark]

2d

(d) The radius of the Earth's orbit around the Sun is:

List the known quantities:

- Gravitational force between Earth and Sun, $F = 3.52 \times 10^{22} \text{ N}$
- Mass of Earth, $m_1 = 6.0 \times 10^{24} \text{ kg}$
- Mass of Sun, $m_2 = 2.0 \times 10^{30} \text{ kg}$

Determine the equation for the radius of the Earth's orbit, r .

$$\bullet \quad F = \frac{Gm_1m_2}{r^2} \Rightarrow r = \sqrt{\frac{Gm_1m_2}{F}} \quad [1 \text{ mark}]$$

Substitute in the values:

- $r = \sqrt{\frac{(6.67 \times 10^{-11}) \times (6.0 \times 10^{24}) \times (2.0 \times 10^{30})}{3.52 \times 10^{22}}}$ [1 mark]
- $r = 1.5 \times 10^{11} \text{ m}$ [1 mark]

[Total: 3 marks]

r is always measured from the **centre** of the two masses that we are calculating the gravitational force for. The radius of orbit of the Earth around the Sun is also measured in this way. You must recognise different wording for the same thing!

To help, look at what variables you have already been given. Although you could use a circular motion equation, we haven't been given the linear speed which could make this difficult. However, since we have been given the gravitational force, we can also use Newton's law of gravitation.

3a

(a) A geostationary orbit is:

- An equatorial orbit / about the equator; [1 mark]
- Where a satellite moves from west to east / in the same direction as the Earth spins; [1 mark]
- With a period of 24 hours / 1 day; [1 mark]

[Total: 3 marks]

3b

(b) M is given by $M = \frac{v^2 r}{G}$ because:

Equate the centripetal and gravitational force:

- $F = \frac{mv^2}{r} = \frac{GMm}{r^2}$ [1 mark]

Cancel and rearrange for mass M :

- $\frac{\cancel{m}v^2}{r} = \frac{G\cancel{m}M}{r^2}$

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- $F = \frac{mv^2}{r} = \frac{GMm}{r^2}$ [1 mark]

Cancel and rearrange for mass M :

- $\cancel{m}v^2 = \frac{GM\cancel{m}}{r^2}$
- $v^2 = \frac{GM}{r}$
- $M = \frac{v^2 r}{G}$ [1 mark]

[Total: 2 marks]

3c

(c) M can also be written as $M = \frac{4\pi^2 r^3}{T^2 G}$ because:

Determine an equation for v in terms of T :

- $v = \omega r = \frac{2\pi r}{T}$ [1 mark]

Substitute this equation into the equation for M in terms of v :

- $M = \frac{v^2 r}{G} \Rightarrow M = \frac{\left(\frac{2\pi r}{T}\right)^2 r}{G}$
- $M = \frac{\frac{4\pi r^2}{T^2} r}{G} = \frac{4\pi r^3}{T^2 G}$ [1 mark]

[Total: 2 marks]

Since satellites travel roughly in a circular path, you will need to be familiar with all the circular motion equations for this section of gravitational fields as well.

3d

(d) The radius of a geostationary orbit around Earth is:

List the known quantities:

- Mass of the Earth, $M = 6.0 \times 10^{24} \text{ kg}$
- Time period of a geostationary orbit, $T = 24 \text{ hours}$
- gravitational constant, $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

Calculate the time period in seconds, T :

- $T = 24 \text{ hours} = 24 \times 60 \times 60$
- $T = 86\,400 \text{ s}$ [1 mark]

Rearrange the mass equation for the radius or orbit, r :

- $r = \sqrt[3]{\frac{MT^2 G}{4\pi^2}}$ [1 mark]

Substitute values into the radius equation:

- $r = \sqrt[3]{\frac{(6.0 \times 10^{24}) \times (86\,400)^2 \times (6.67 \times 10^{-11})}{4\pi^2}}$ [1 mark]
- $r = 4.2 \times 10^7 \text{ m}$ [1 mark]

[Total: 4 marks]

